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Key Points:

- An original numerical framework assesses equatorial waves' contributions to a record-breaking extreme precipitation event in Central Sahel
- Tropical disturbances and Equatorial Rossby waves are pivotal in driving this extreme precipitation event

Supporting Information:

Supporting Information may be found in the online version of this article.

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Exploring the Influence of Equatorial Waves on a Record-Breaking Extreme Precipitation Event in Central Sahel: Insights From Convection Permitting Simulations

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Abstract The floods of 01 September 2009 in Ouagadougou, Burkina Faso, were among the most severe in the Central Sahel's history, highlighting the region's persistent flood risk. This extreme precipitation event (EPE) was influenced by various atmospheric drivers, including Convectively Coupled Equatorial Waves (CCEWs). This study uses the Meso-NH numerical model to assess the individual impacts of CCEWs on the 2009 EPE. Results indicate that Tropical Disturbances (TDs) and Equatorial Rossby waves (ERs) significantly affect the system's propagation and intensity, with TD removal causing a northward shift and ER removal reducing precipitation due to upstream drying. Removing both TDs and ERs eliminates the EPE entirely in Ouagadougou, while other waves have minimal impact. These findings emphasize the crucial role of TD and ER interactions in triggering EPEs in the Central Sahel and highlight the importance of accurately representing these waves in forecasting models.

Plain Language Summary In 2009, Ouagadougou, Burkina Faso, faced devastating floods due to heavy rainfall, making it one of the worst disasters in the country's history. This study aims to understand the factors behind this extreme precipitation event (EPE). Using advanced numerical simulations and observational data, researchers investigate the role of different atmospheric phenomena, including equatorial waves, in causing these floods. We find that certain types of equatorial waves, such as TDs and ER waves, significantly influenced the intensity, speed and location of rainfall. Removing these waves from the simulations leads to changes in the precipitation patterns, highlighting their importance. These results provide valuable insights to improve our understanding of extreme weather events and could help in developing better strategies for managing and preparing for such disasters in the future.

1. Introduction

On 1 September 2009, Ouagadougou experienced an exceptional 10-hr rainfall accumulation exceeding 260 mm, producing catastrophic flash flooding and setting the historical rainfall record with a >100 – yr return period (Engel et al., 2017; Lafore et al., 2017). As climate change intensifies hydrometeorological extremes, such events pose increasing risks for rapidly urbanizing Sahelian cities (IPCC, 2023).

This record-breaking extreme precipitation event (EPE) has been documented in a few studies. Lafore et al. (2017) identified that the EPE was characterized by a co-occurrence of multiscale drivers: at interannual scale, associated with a La Niña phase of El Niño Southern Oscillation (ENSO), and at intraseasonal scale, marked by a strong activity of convectively coupled equatorial waves (CCEW). In fact, a train of three TDs, the so-called African Easterly waves, was embedded in a large-scale moist envelope in the northern Sahel, later identified by Peyrillé et al. (2023) as an ER wave. The ER played a key role by moistening the atmosphere north of Ouagadougou while the TD provided an intense relative vorticity, above the 99.9th percentile (1979–2014) of TD's vorticity (Engel et al., 2017). A Kelvin wave (KW) increased convection to the east of Ouagadougou before the occurrence of the EPE while a weak magnitude Madden-Julian Oscillation (MJO) enhanced convection north of 10°N.

Peyrillé et al. (2023), based on a statistical analysis, demonstrated that CCEWs are key in the occurrence of EPE in the central Sahel. They revealed that the ER dominates on a large scale, providing a moist and convectively active anomaly over the northern Sahel, while TDs are the most frequent on the synoptic scale. The MJO

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contributes to upper-level divergence anomalies, and the KW increases convection during EPEs. They also showed that some preferential co-occurrences of CCEWs lead to an increased probability of EPEs. The most favorable configurations always involve a convectively active TD together with a larger-scale wave such as an ER and/or KW and, to some extent, the MJO. Applying their calculations to the 2009 Ouagadougou EPE event, we find that it corresponds to an ER + TD configuration (not shown), with the KW and MJO being of weak magnitude compared to each wave standard deviation at the time and location of the EPE.

The moist environment surrounding the convective system and the intense vorticity have been identified as key features of the situation. Previous studies (Engel et al., 2017; Lafore et al., 2017; Peyrillé et al., 2023) have relied on in situ rainfall measurements, satellite-derived products (such as Climate Hazards Group InfraRed Precipitation with Station data, Tropical Rainfall Measuring Mission, or Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks - Climate Data Record), and reanalysis data sets (e.g., ERA-Interim) to investigate the multiscale drivers of the 2009 Central Sahel EPE. Although these studies highlighted the role of tropical waves such as ER and TD in moistening the environment and enhancing vorticity, they did not quantify the specific contribution of each wave to convective evolution or precipitation intensity. Although one might anticipate that the ER and TD are the leading drivers, no modeling-based evidence has yet been provided to clarify how CCEWs impact and shape a regular convective system into a convective system leading to an EPE.

To fill this gap, the present study proposes a novel approach using a convection-permitting numerical simulation framework driven by operational meteorological analyses. The objective of this article is to provide evidence of the impact of each wave on the convective system and how they helped a record-breaking EPE to form over Ouagadougou. To that aim, we design a series of CCEW-removal sensitivity experiments for which the contribution of each CCEW in the initial state and on the lateral boundaries have been removed. The use of a convection-permitting model has been shown to be efficient in reproducing this EPE (Beucher et al., 2019) while the CCEW-removal sensitivity-experiment strategy has been used successfully to quantify the role of CCEW on a tropical cyclone (Ching et al., 2015) but has not yet been applied to the analysis of EPEs.

This is, to our knowledge, the first numerical sensitivity framework to attribute a Central Sahel EPE to individual CCEW components. Section 2 presents the methodology and data used, Section 3 discusses the results, and the final section covers the conclusion and prospects.

2. Data Sets and Methods

2.1. Observations and Analyses Data Sets

We use the National Aeronautics and Space Administration (NASA) Integrated Multi-satellite Retrievals for Global Precipitation Measurement (IMERG) half-hourly precipitation (0.1°, Huffman et al. (2015)), hourly Meteosat Second Generation (MSG) 10.8 μm brightness temperature (BT; 0.1°), and National Ocean and Atmosphere Administration (NOAA) daily Outgoing Longwave Radiation (OLR; 2.5°; Lee & Program, 2011). Large-scale moisture and circulation are characterized using 3-hourly ERA5 precipitable water (PW) and winds at 925 and 700 hPa (0.25°; Hersbach et al. (2020) and compared to the numerical simulations.

2.2. Meso-NH Convection Permitting Model

Simulations are performed with the nonhydrostatic Meso-NH model (Lac et al., 2018) at 2.5 – km grid spacing and 62 vertical levels up to 26 km. Runs start at 00 UTC 31 August 2009 and last 48 hr over the domain (red rectangle in Figure 1). Initial and lateral boundary conditions are taken from 6-hourly ECMWF operational analyses (0.1°; 91 levels). Physics options are summarized in Text S1 in Supporting Information S1.

2.3. Isolating Contributions From CCEWs

To isolate individual equatorial-wave signals, we apply wavenumber–frequency filtering (Roundy & Frank, 2004; Wheeler & Kiladis, 1999). We filter 6-hourly ECMWF operational analyses (winds, temperature, humidity) on all model levels to generate filtered initial and boundary conditions for the sensitivity experiments. For diagnostics, we similarly filter ERA5 700-hPa winds and National Oceanic and Atmospheric Administration (NOAA) outgoing longwave radiation (OLR). Spectral domains follow Peyrillé et al. (2023) and are provided in Text S2 and Figure S1 in Supporting Information S1.

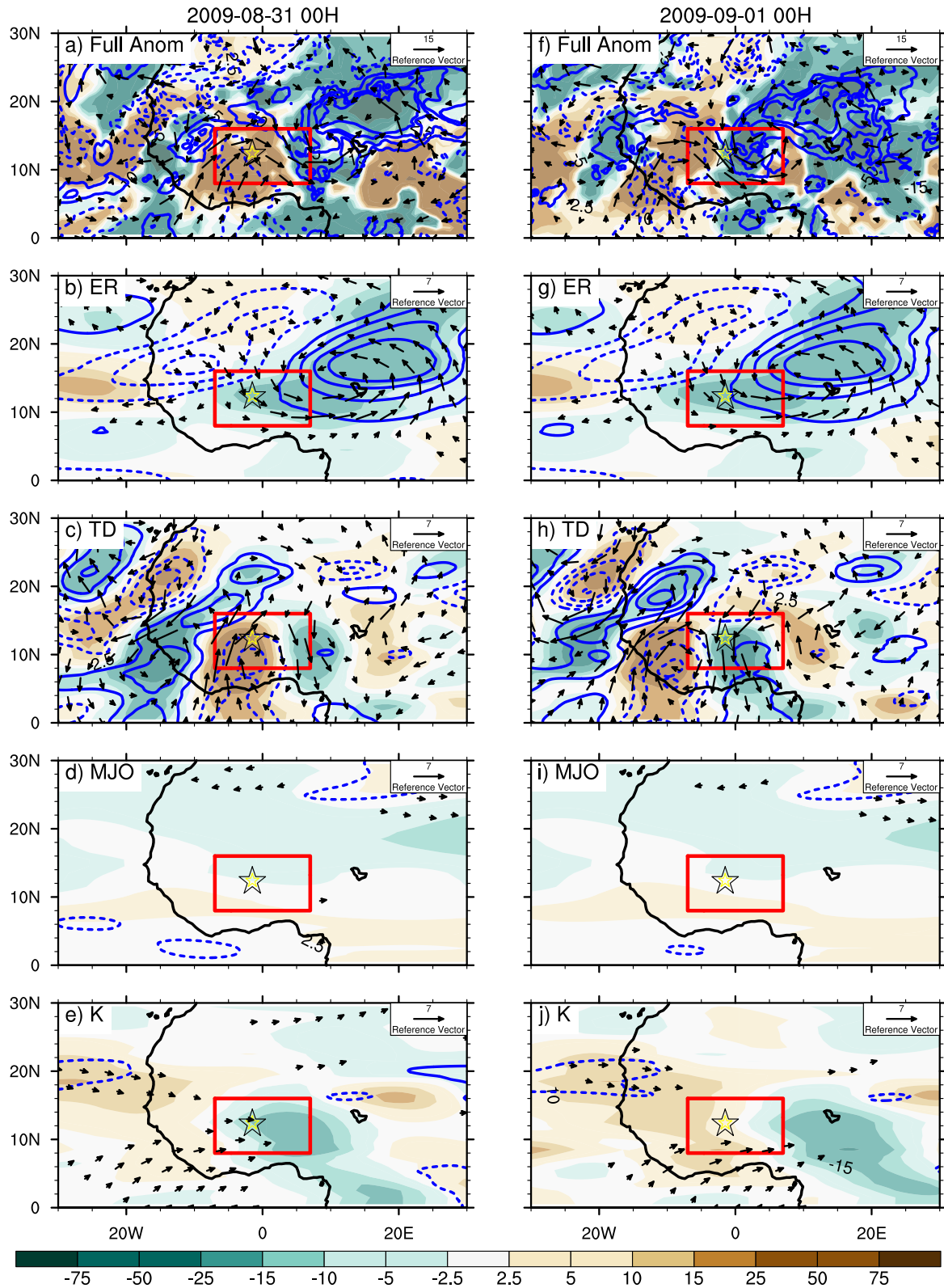


Figure 1.

2.4. Design of CCEW-Removal Sensitivity Experiments

In total, six simulations of 48 hr have been run. The control experiment (CTL) is performed with the raw unfiltered analysis data for initial fields and lateral boundary conditions. Then 5 CCEW-removal sensitivity experiments explore the sensitivity of the simulated rainfall and convection to alterations of the initial state and lateral boundary conditions. A given CCEW-removal sensitivity experiment is built by removing the contribution of this given wave on horizontal winds, temperature, specific humidity and surface geopotential obtained after wavenumber-frequency filtering for both the initial state and the time-evolving lateral boundary conditions. The details of the methodology are given in Text S3 and Figures S2–S5 in Supporting Information S1. Experiments NoTD, NoER, NoKW, NoMJO thus refer to the suppression of the respective signature of TD, ER, KW and MJO. A supplementary experiment, No(ER + TD), is realized where both the contributions of ER and TD are removed. We assess the changes in atmospheric balances associated with the removal of a given wave by calculating the domain-averaged ratio of ageostrophic wind against geostrophic wind (Table S1 in Supplementary Information S1). We found that this measure of atmospheric balance shows slight variations between 0.9 and 1 among the different experiments. The largest imbalance occurred when the Equatorial Rossby (ER) wave was removed, particularly in the No(TD + ER) experiment, highlighting its key role in maintaining large-scale balance.

3. Results

3.1. Identification of CCEWs Contributions During the EPE

Figure 1 presents observed anomalies from the climatology of precipitable water, OLR and wind at 700 hPa for the day before the EPE, on 31 August at 00 UTC and at the time of the EPE, on 01 September at 00 UTC. The full anomalies (Figures 1a and 1f) show that the EPE event is characterized by the presence of a broad moist and convective anomaly on the northeastern Sahel, to the northeast of Ouagadougou which barely moves. Focusing on Ouagadougou, a dry anomaly prevails on 31 August which turns moist on 01 September associated with a wind change from southwesterly to northwesterly. The decomposition of this full anomaly in contributions from the different CCEW indicates that the contributions of TD (Figures 1c and 1h) and ER waves (Figures 1b and 1g). A quantitative decomposition for a $1^\circ \times 1^\circ$ box centered on Ouagadougou is provided in Table S2 in Supporting Information S1 and confirms the dominant contributions of TD and ER. In particular, the large moist anomaly comes from the ER, while the TD provides vorticity and humidity at a smaller scale. Contributions from the MJO (Figures 1d and 1i) are very weak if not negligible. The KW contribution shows a convectively active signal east of Ouagadougou one day before the EPE (Figures 1e and 1j) with anomalous westerlies south of Ouagadougou. The convective KW anomaly moves eastward leaving a convective inactive anomaly prevailing at Ouagadougou on 01 September. The ER contribution precedes the event, creating the humid conditions northeast of Ouagadougou, while the TD contribution appears to influence the northward flux and the vortex intensity. The KW might increase convection one day before further to the east but seems unfavorable to convection at the time of the EPE. In the classification of Peyrillé et al. (2023), the EPE is indeed an ER + TD case with no active MJO and a convectively inhibited phase of KW. This suggests that the EPE could have been even stronger if these last two CCWs had been active. This analysis motivates the use of the Meso-NH model to better demonstrate the effects of each wave.

3.2. Sensitivity of the Simulated EPE to CCEWs Removal

Figure 2 shows the average situation over 24 hr (from 31 August at 12 UTC to 01 September at 12 UTC) surrounding the peak of the Ouagadougou event. This 24-hr average allows us to adhere to the classical definition of extreme rainfall events used in the literature and better appreciate the extreme nature of the Ouagadougou case (Peyrillé et al., 2023; Sanogo et al., 2022). Indeed, IMERG observations show that the maximum rainfall occurs around Ouagadougou with an intensity of 100 mm (Figures 2a and 2b), while MSG observations indicate that the coldest part of the system (i.e., the region of minimum BT) is slightly larger and located slightly farther east, with

Figure 1. Top row: Full Anomalies for NOAA Outgoing Long-wave Radiation (shading, W m^{-2}), ECMWF operational analysis 700-hPa wind (vectors, m s^{-1}) and Precipitable Water (blue contours, contours every 2 kg m^{-2}) on (a) 31 August at 00 UTC and (f) 01 September 2009 at 00 UTC. (b–j) Same as (a, f) except for the respective contributions of each convectively coupled equatorial wave: (b, g) Equatorial Rossby wave (ER), (c, h) Tropical Disturbance (TD), (d, i) Madden-Julian Oscillation (MJO) and (e, j) Kelvin wave (K). Contour every 2 kg m^{-2} except in (a, b) where the contour is every 8 kg m^{-2} . The indicates the location of Ouagadougou, while the red rectangle represents the simulation domain used in this study.

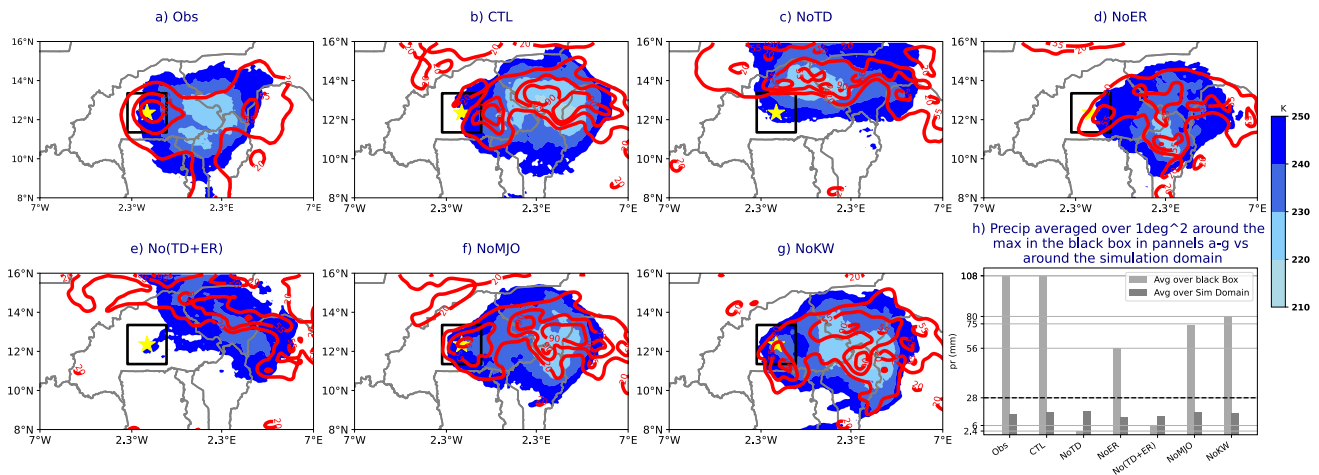


Figure 2. Daily average between 31 August 12 UTC and 01 September 12 UTC of BT (K, shading) and rainfall (mm, contours at 20, 55 and 90 mm) for: (a) observations (BT, MSG and rainfall, IMERG), (b) CTL experiment, (c) NoTD simulation, (d) NoER simulation, (e) No(TD + ER) simulation, (f) NoMJO simulation, and (g) NoKW simulation. Panel (h) shows the rainfall daily accumulation average in the square degree centered on the rainfall maximum around Ouagadougou (black box in panels a–g). The yellow stars stand for the position of Ouagadougou.

relatively lower rainfall amounts (Figure 2a). Regarding Meso-NH CTL simulation, the EPE around Ouagadougou is well captured, although slightly shifted toward the east (Figure 2b). Meso-NH demonstrates a good distribution in BT but exhibits excessive precipitation within the cold part of the system compared to IMERG precipitation. These discrepancies can however either be interpreted as a limitation of satellite-derived precipitation product like IMERG to produce high rainfall (Satgé et al., 2019) or as a limitation of state-of-the-art convective permitting models.

Here we use the numerical model as a virtual laboratory in order to quantify the effect of each individual wave with the CCEW-removal sensitivity simulations. The maximum precipitation is entirely shifted northward in simulation noTD (Figure 2c), leaving it within the cold BT area. Without ER, there is no significant change in the system location, but the magnitudes of the rainfall and BT patterns drastically decrease (Figure 2d). In the No (TD + ER) simulation (Figure 2e), the amplitude of precipitation and BT significantly weakens. The circular shape of the system becomes distorted, closer to that of a squall line and the system's core is deflected northward. Therefore, the effects of TD and ER noted respectively in Figures 2c and 2d, appear to accumulate. In contrast to those three experiments, no significant changes are found when eliminating the MJO (Figure 2f) and KW (Figure 2g) forcings. We then compare in Figure 2h the simulated rainfall daily accumulation average in the square degree centered on the rainfall maximum around Ouagadougou (black box in panels a–g of Figure 2), which is largely above the 99th percentile of the annual distribution (here defined as 28 mm/day), a usual threshold used to define EPE (Peyrillé et al., 2023; Sanogo et al., 2022). The CTL simulation shows a daily accumulation comparable to observations around Ouagadougou, and the daily accumulation is halved in noER, falling below the 28 mm threshold only in the simulations noTD and no(ER + TD). To further assess whether these reductions result from weaker convection or from a displacement of the convective system, we also compared the rainfall averaged over the entire simulation domain (Figure 2h). This additional diagnostic shows that removing the equatorial Rossby wave (NoER) induces the largest decrease in the domain-mean rainfall, whereas in NoTD, the domain-mean rainfall remains close to the control and observations. However, rainfall inside the Ouagadougou-centered box drops sharply, indicating that the EPE does not dissipate but is displaced northward, leaving Ouagadougou dry while maintaining strong rainfall nearby.

Therefore, one can conclude that TD and ER are the CCEWs that present the most significant influence on the Ouagadougou EPE. The additional analyses provided in Supporting Information S1 confirm that ER mainly controls the overall rainfall intensity, whereas TD primarily governs the system's displacement and thus the local occurrence of the EPE. Without TD, the system does not pass over Ouagadougou, resulting in no occurrence of EPE, and without ER, the intensity of the EPE is halved, while their effects accumulate.

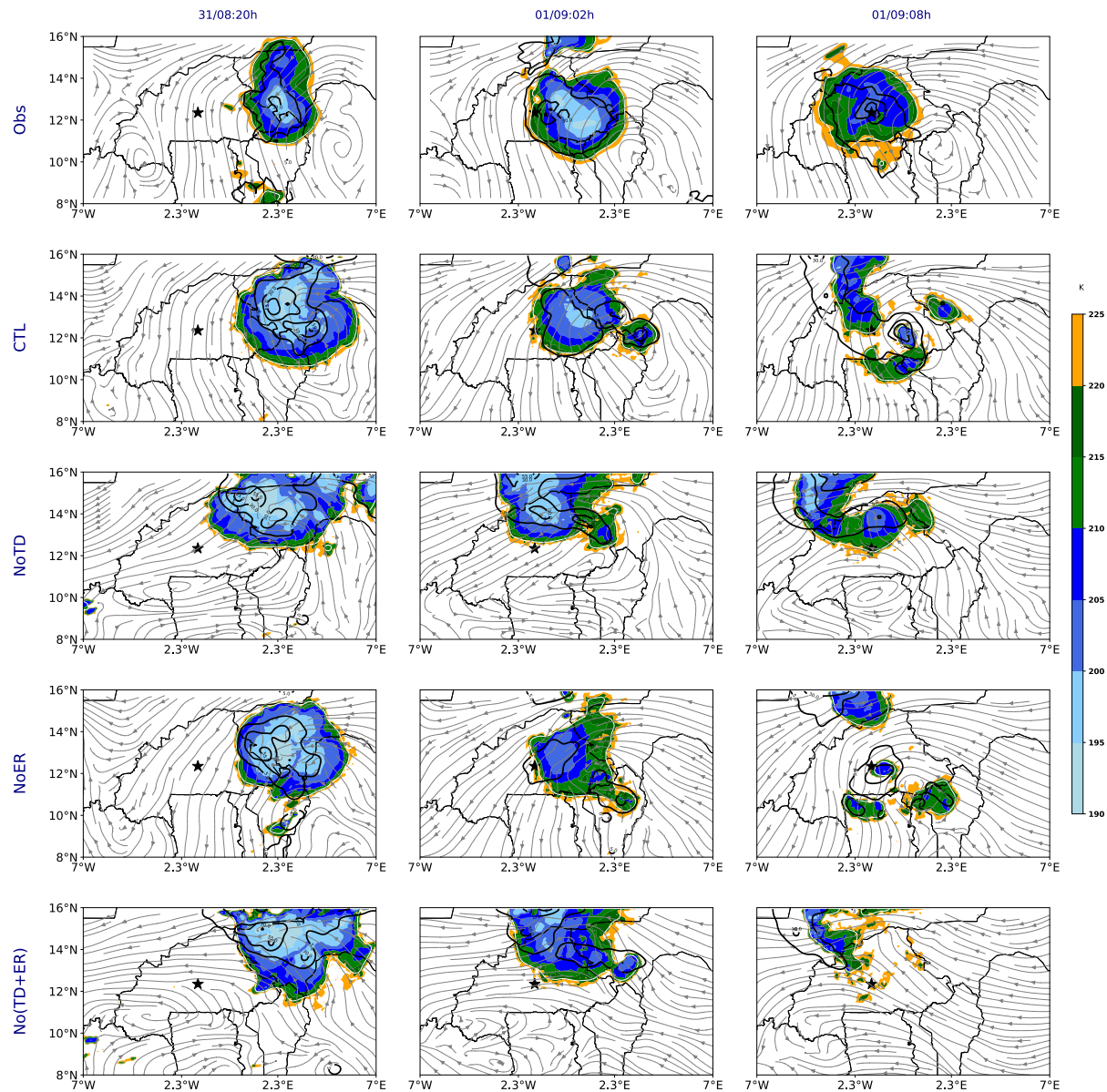


Figure 3. Brightness temperature (BT) in color, cumulative rainfall over the previous 3 hr in black contours (with intervals of 25 mm), horizontal wind at 700 hPa in streamlines at three selected times: 31 August at 20 UTC in the first column, 01 September at 02 UTC in the second column, and 01 September at 08 UTC in the third column. Observations (BT from MSG, rainfall from IMERG, and wind from ERA5) are represented in the first row, CTL experiment in the second row, simulations noTD in the third row, noER in the 4th row, no(TD + ER) in the 5th row. The thin white contour represents the 220 K BT threshold and the black star indicates the position of Ouagadougou.

After analyzing the daily average situation of the EPE, we examined the evolution of the EPE throughout the day. Figure 3 illustrates the system's evolution at three key moments (before it reaches Ouagadougou, at the time the system enters Ouagadougou, and when the system transitions from a more circular structure to a squall line in the CTL simulation). The observations highlight the circular structure of the convective system, especially from 02 UTC on 01 September and onwards, with the maximum rainfall consistently observed in the coldest part of the system, located on the northern edge of the cyclonic vortex accompanying it. The CTL simulation accurately captures this organization during the growth and maturity phases of the system, despite a premature deformation of the system and an earlier circular shape. The phasing of precipitation, cloudiness, and winds are also comparable, leading to an overall good agreement between the CTL experiment and observations.

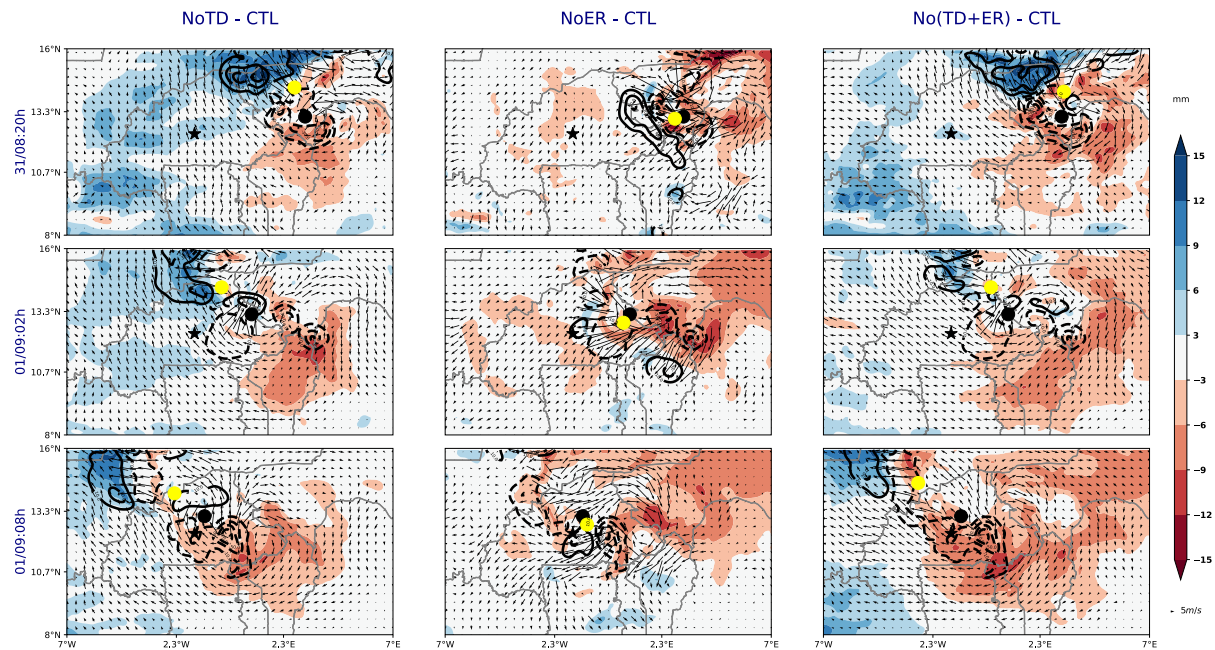


Figure 4. Left: NoTD-CTL differences of PW (kg m^{-2} , shading), 3-hr cumulative rainfall (mm, contours every 20 mm) and 925 hPa horizontal wind (ms, vectors) at three key instants (31/08 12 UTC, 01/09 02 UTC, 01/09 08 UTC). Middle and right: same as Left except for NoER-CTL and No(TD + ER)-CTL, respectively. The rainfall accumulation is computed over the past 3 hr, Black and yellow dots represent the barycenter of the convective system of the CTL and sensitivity simulations, respectively. The black star denotes the position of Ouagadougou.

In the sensitivity experiments, in noTD, the system deviates northward. This situation arises from the weakening of the TD northwesterly flux associated with the cyclonic vortex shown in Figure 1f and stands as a prominent characteristic of the EPE of Ouagadougou (Figures 1f and 1h). The system propagation also appears faster in noTD, due to the reduction in its cyclonism. In noER, the southward flux remains intact, and the system latitudinal position remains unchanged. However, the surface of the coldest part of the system weakens more rapidly compared to the CTL simulation and observations. This indicates a very rapid decrease in the system intensity in noER simulation. Removing both TD and ER results in the disappearance of the southward flux, causing the system to shift northward and its intensity to decrease more rapidly than CTL experiment. This suggests that the influence of these two waves cumulates.

The analysis of the sensitivity experiments at these three key instants confirms that the ER provides humidity, while the TD tends to slow and deflect the system. This is even more evident in the differences between the sensitivity experiments and CTL shown in Figure 4.

In the noTD simulation, a wet anomaly is observed in the northwest, contrasting with dry anomalies in the southeast (Figure 4 left). The centroid of the observed convective system, calculated as is also positioned northwestward compared to that of the CTL simulation, suggesting a faster system along a trajectory further oriented northward. The longitudinal speed of the centroid in the noTD simulation between 31 August at 20 UTC and 01 September at 08 UTC is 13.23 m/s, faster than the CTL simulation speed of 11.1 m/s. The suppression or decrease of the northwesterly wind, as mentioned in Figure 3, corresponds with a dry anomaly and a southerly difference to the southeast of the system, highlighting the significant impact of the TD. This suggests that the TD cyclonic circulation contributed to moistening the area and maintaining moisture supply to the convective system, even when it was positioned westward of the main humid anomaly associated with the ER (see Figures 1c and 1i). Thus, the TD not only steers the system's trajectory southward but also provides moisture to the convective system from the ER humid envelope northeast of the system, preventing the formation of a moist circular system at Ouagadougou in its absence.

In the noER simulation, the propagation speed and position of the convective system core remain relatively unchanged, with a longitudinal speed of 9.69 m/s between 31 August at 20 UTC and 01 September at 8 UTC (Figure 4 middle). However, a broad negative precipitable water anomaly is noted northeast of the system, with a

less pronounced wind signature than in the TD removal scenario, up until 01 September at 06 UTC. At this point, a large-scale anticyclonic anomaly is observed, corresponding to anomalous drying in the northeastern part of the domain. The drying associated with ER removal weakens the system but does not inhibit its development into a high-precipitation storm, as evidenced in Figure 2.

Finally, the removal of both TDs and ERs leads to the complete disappearance of the EPE system, accompanied by dry anomalies both to the northeast and southern part of Ouagadougou (Figure 4 right). The combined effects of removing both TD and ER show a more rapid longitudinal speed of 16.06 m/s between 31 August at 20 UTC and 01 September at 08 UTC, significantly higher than that of the CTL simulation. This result illustrates that the interactions between these drivers at different scales (ER at the large scale and TD at the synoptic scale) constructively enhance deep convection in the Ouagadougou area.

4. Conclusions

In this study, we numerically investigate the intricate relationship between CCEWs and EPE in Central Sahel. By removing the individual contributions of CCEWs within convective permitting numerical simulations, we demonstrate that ER and TD are the two interacting drivers that were causing the occurrence of the 01 September 2009 EPE.

Our findings highlight that the ER plays a crucial role in maintaining moisture supply in the northern area of the convective system sustaining convective activity. However its removal still leads to an EPE rainfall accumulation. In contrast, the intense TD is critical to the occurrence of the EPE in Ouagadougou by directing the trajectory of the convective system toward the city and most importantly by extracting humidity from the ER moist envelope, allowing a moist circular-shaped system to develop. The combined influence of these equatorial waves amplifies the intensity of precipitation events in Ouagadougou, as evidenced by the total disappearance of the event when both TDs and ERs are removed.

A major finding here is that it is the interaction of these two waves that allowed an EPE to occur in Ouagadougou. If the ER is removed, the TD can not extract enough humidity to form a circular system and a squall line is observed. If the TD is removed, the convective system remains in the ER envelope, without a synoptic forcing. Regarding results by Peyrill   et al. (2023) who found the combination of ER + TD to be one of the most frequent and effective to lead to EPE in Central Sahel, we argue that the mechanisms found here can be extended to Central Sahel EPEs. In particular the monitoring of a TD and convective system developing on the western side of an ER moist anomaly is a key feature of this EPE that should be monitored by Sahelian forecasters to be the basis of early warning systems.

Moreover Engel et al. (2017) pointed out that the TD vorticity was above the 99.9th percentile during this EPE. A remaining question is whether or not an extremely intense TD was needed to generate this EPE or if a regular TD within an ER is sufficient. Peyrill   et al. (2023) found that half of EPEs occurrences are associated with a TD but their study was based on OLR and not on vorticity which appears to be a key dynamical aspect. Future studies should investigate the dynamical signature of TDs leading to EPE.

Understanding how these waves interact and amplify each other's impacts could provide valuable insights for improved forecasting and early warning systems. It also opens perspectives for investigating how climate change may alter the thermodynamic environment in which these wave interactions produce EPEs in the Central Sahel and beyond.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All data used in this study are publicly and freely available. The simulations were performed with version 5.5 of the Meso-NH model (Lac et al., 2018). The Integrated Multi-satellitE Retrievals for GPM (IMERG) data are described by Huffman et al. (2015). The MSG2 10.8 μm brightness temperature data are based on Meteosat Second Generation observations (Schmetz et al., 2002) and were obtained through the ICARE Data and Services

Center. The NOAA interpolated OLR data are described by Lee and Program (2011). The ERA5 reanalysis is described by Hersbach et al. (2020). The ECMWF operational analysis data used in this study were retrieved through the ECMWF MARS archive.

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